



SWARM

B-Lite → B-Full · Autonomous Infrastructure Evolution v2

**The objective is not building magic immediately.
The objective is progressively reducing human configuration until the
system feels autonomous.**

01 · What SWARM Actually Is

SWARM is not a drone company.

SWARM is autonomous coordination infrastructure.

The drones are replaceable. The coordination layer is the product.

Core moat:

- orchestration
- telemetry
- mapping
- environmental understanding
- distributed coordination

02 · Definition of B-Lite

B-Lite is constrained autonomous infrastructure.

Characteristics:

- plug-and-play onboarding
- minimal setup
- autonomous patrol logic
- scheduled territory scans

- anomaly verification
- partial self-mapping

The system operates in constrained environments where autonomy can realistically succeed.

03 · Constrained Environments

Initial deployments should avoid extremely complex environments.

Best early environments:

- vineyards
- open rural properties
- resorts
- agricultural land
- low-density territories

Worst early environments:

- dense urban areas
- highly occluded properties
- chaotic environments

04 · Persistent Awareness

The core objective is not random drone patrol.

The objective is persistent territorial awareness.

SWARM combines:

- rotating aerial coverage
- periodic scans
- environmental sensing
- anomaly verification

The system progressively increases awareness continuity over time.

05 · Rotational Coverage Logic

Continuous drone hovering is not economically viable initially.

Instead:

- drones rotate
- units alternate charging cycles
- patrol frequency adapts dynamically
- coverage prioritizes high-risk zones

SWARM optimizes awareness while minimizing energy usage.

06 · What B-Lite Actually Does

B-Lite does not fully protect every territory.

B-Lite creates:

- territorial awareness
- wildfire early detection
- anomaly verification
- environmental visibility
- reduced response time

The first value is awareness and verification. Not total automation.

07 · Phase 0 · Foundation

0–3 months.

Focus:

- ROS2
- drone SDKs
- simulations
- telemetry systems
- path planning

- computer vision

Output: SWARM OS simulation environment.

08 · Phase 1 · First B-Lite Prototype

3–6 months.

Build:

- dock v0
- 2–4 drones
- scheduled patrol system
- autonomous launch/return
- simple anomaly verification

Human configuration still exists.

09 · Phase 2 · Territory Understanding

6–12 months.

SWARM progressively learns:

- geometry
- obstacles
- safe flight zones
- visibility gaps
- optimal patrol routes

Manual setup begins decreasing.

10 · Phase 3 · Patrol Intelligence

12–18 months.

Build:

- adaptive patrol logic
- battery rotation systems
- environmental risk scoring
- dynamic route generation
- anomaly prioritization

Coverage becomes contextual instead of static.

11 · Phase 4 · Persistent Awareness Layer

18–30 months.

Introduce:

- rotating aerial awareness
- optional tethered node
- cooperative scanning
- hybrid sensing

SWARM transitions from periodic awareness toward persistent awareness.

12 · Phase 5 · Swarm Coordination

30–42 months.

The network becomes truly distributed.

Capabilities:

- mission allocation
- cooperative verification
- autonomous sequencing
- multi-unit coordination

13 · Phase 6 · Reliability

42–54 months.

Main focus:

- false positive reduction
- weather resilience
- maintenance reduction
- autonomous recovery
- operational stability

Reliability matters more than cinematic autonomy.

14 · Definition of B-Full

B-Full does not mean zero humans.

B-Full means: minimal required human configuration.

User flow:

- receive SWARM kit
- place dock
- open app
- define broad territory

SWARM handles the remaining operational logic autonomously.

15 · Physical Constraints

Real-world constraints remain critical:

- weather
- battery limitations
- occlusion
- noise
- connectivity
- maintenance

SWARM succeeds only if autonomy remains operationally realistic.

16 · Final Strategic Principle

The company is the progressive reduction of operational complexity.

At the beginning: humans heavily configure the system.

At the end: SWARM largely configures itself.

That transition is the roadmap.

Many units. One intention.